



Exploring therapeutic targets with the HMM-SA structural alphabet: Methods, tools, and application to HIV-2 protease



Anne-Claude Camproux^{*}, Marine Baillif, Léa Dufay, Leslie Regad^{**}

Université Paris Cité, CNRS, Inserm, Unité de Biologie Fonctionnelle et Adaptative, F-75013, Paris, France

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ABSTRACT

The rapid expansion of available three-dimensional protein structures—derived from both experimental techniques and bioinformatic predictions—offers unprecedented opportunities for drug discovery, particularly for targets that have historically been difficult to characterize. However, the effective analysis of these increasingly complex and voluminous structural datasets remains a major challenge. Efficient representations of protein conformations are essential to facilitate large-scale comparison, structural classification, and functional interpretation in therapeutic contexts. The concept of structural alphabets, introduced by Pr S. Hazout in 1999, provides a robust and scalable framework to represent local protein backbone conformations using a limited set of recurring structural motifs. This representation enables a one-dimensional encoding of three-dimensional protein structures that retains essential geometric features beyond secondary structure, while allowing systematic and interpretable analyses. In this review, we focus on HMM-SA, a structural alphabet constructed using a hidden Markov model. HMM-SA defines 27 structural motifs, including 18 regions specifically dedicated to loops, and captures the statistical dependencies between them. We present a detailed overview of the HMM-SA framework, and of the computational tools derived from this structural alphabet, developed to explore protein function, conformational variability, and the structural determinants of molecular recognition. The utility of HMM-SA is illustrated through a case study on HIV-2 protease (PR2), a critical enzyme in antiretroviral drug development. By analyzing PR2 structural asymmetry, ligand-induced conformational changes, and mutation-driven alterations, we highlight the ability of HMM-SA-based methods to identify key structural features involved in ligand specificity and resistance mechanisms, thereby advancing therapeutic target analysis.

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^{*} Corresponding author.

^{**} Corresponding author.

E-mail addresses: anne-claude.camproux@u-paris.fr (A.-C. Camproux), regad@u-paris.fr (L. Regad).

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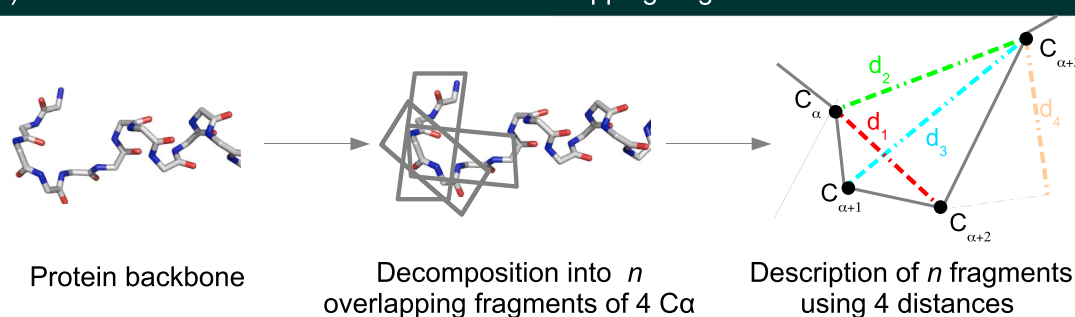
1. Introduction

The characterization and description of three-dimensional (3D) protein structures constitute a fundamental challenge in the domain of structural bioinformatics, particularly within the context of drug design. Accurate characterization of protein conformations is essential for elucidating their biological functions and for optimizing the rational design of therapeutic compounds. However, the analysis and characterization of protein structures poses a substantial computational challenge due to their 3D complexity. To address this, several approaches have been developed, including structural fragment libraries, which enable the decomposition of complex structures into more easily analyzable recurrent elements [1–9]. These libraries have been developed for a variety of purposes, including protein structure reconstruction [1,4], local structural segment prediction [3], and fine-grained conformational space analysis [10,11]. While geometrically accurate, these approaches generally overlook the contextual organization of structural fragments along the polypeptide chain. To address this limitation, the concept of structural alphabet was introduced. It aims to discretize the 3D conformational space of proteins into an ordered sequence of structural letters (SLs), each representing a recurrent local conformation. In contrast to classical fragment libraries, structural alphabets incorporate logical rules governing fragment transitions, thus introducing a sequential dimension into structural representation. The first developments in this direction date back to Zhang et al. (1993) [12], who proposed the concept of Structural Building Blocks, followed by other notable alphabets such as Protein Blocks [9], Conformational Letters [13] (Wang and Zheng, 2008), and M32K25 [14]. In this context, HMM-SA (Hidden Markov Model – Structural Alphabet), introduced by Camproux et al. (1999, 2004), represents a significant methodological advance. It was one of the first structural alphabets to explicitly integrate the joint learning of fragment geometry and transition logic using a machine learning approach [8,15]. HMM-SA was trained using hidden Markov chain models to simultaneously learn the shape of 4-residue fragments of proteins described by their local conformation and the logic of their connections (Fig. 1A). This learning has resulted in (i) an alphabet of 27 4-residue fragment classes, named structural letters, where each structural letter is labeled [A–Z,a] and (ii) the transition matrix between the letters (Fig. 1B, [8,15]). As shown in Fig. 1, HMM-SA encodes 3D structures into sequences of structural letters, each representing the folding of a 4-Calpha (C α) fragment. HMM-SA offers several advantages: its 27 structural letters enable a simplified yet precise description of local structures, particularly of loop regions with 18 dedicated letters [15]. By transforming 3D structures into sequences of structural letters, it facilitates large-scale structural comparisons without the need for optimal

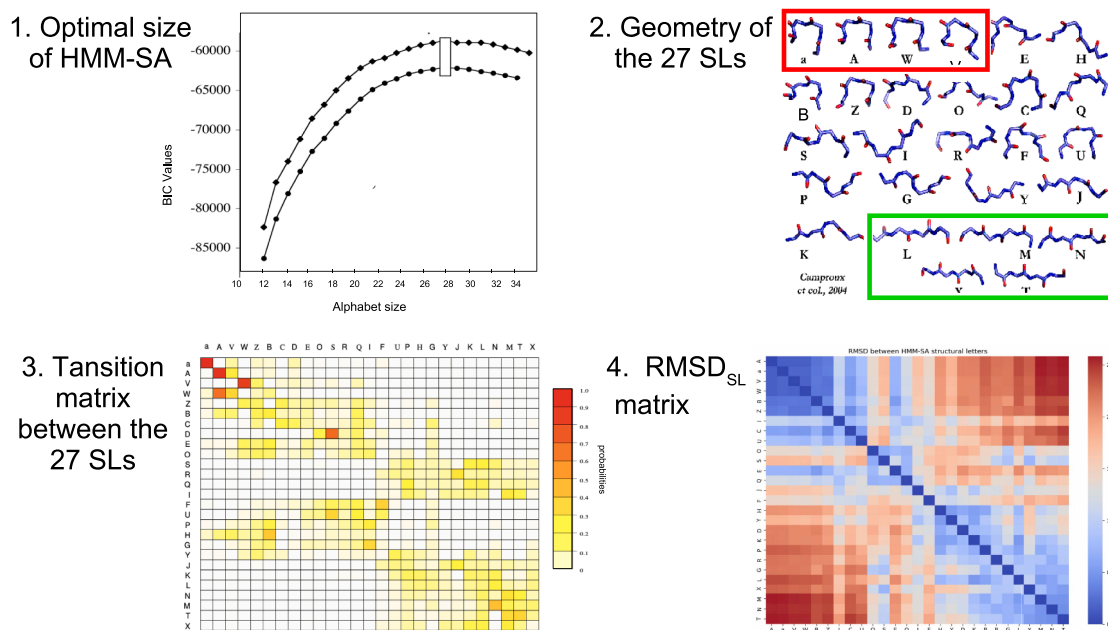
superposition [16]. These features make it particularly well suited to the study of protein conformational diversity and the identification of conserved structural signatures. Interest in this structural alphabet has been highlighted in several reviews, including those by Regad et al. (2008) and Reynès et al. (2011) [16,17], who laid the theoretical and methodological foundations for this approach.

Since the creation of the Protein Data Bank (PDB) [64], the number of experimentally resolved protein structures has grown exponentially. This growth offers the opportunity to explore protein conformational diversity by comparing multiple structures of the same protein obtained under different experimental or biological conditions. In parallel, recent advances in protein structure prediction—particularly the introduction of AlphaFold2 [18]—have revolutionized our ability to predict accurate 3D protein structures directly from amino acid sequences. These methods have significantly advanced the realization of Anfinsen's hypothesis and overcome the complexity implied by the Levinthal paradox. Despite this progress, understanding how protein structures vary across different functional states, respond to mutations, or adapt to the presence of ligands and protein partners remains a major challenge in structural biology. While predictive models such as AlphaFold2 provide static representations of protein folds, they do not capture the full range of conformational variability associated with biological activity. Yet, this structural plasticity is often crucial to function. Proteins may adopt distinct conformations depending on their interaction partners, post-translational modifications, or environmental context. Mutations—whether naturally occurring or acquired—can induce subtle or profound conformational changes that affect protein stability, activity, or binding affinity. Studying these structural variations is essential for several key objectives in molecular biology and biomedicine. These include deciphering the molecular mechanisms underlying drug resistance, characterizing allosteric regulation, and improving the rational design of therapeutic molecules. To support such analyses, researchers need high-resolution, interpretable tools capable of detecting and quantifying local conformational changes across multiple protein structures. These tools must not only describe static geometry but also capture variability in a systematic, scalable, and biologically meaningful way. In this context, structural alphabets such as HMM-SA provide a powerful and complementary approach to deep learning-based methods. HMM-SA discretizes protein backbone geometry into a sequence of structural letters, offering a simplified yet precise representation of local conformation. This approach supports large-scale comparison of protein structures and enables the detection and quantification of conformational diversity.

Recent developments using HMM-SA have led to the creation of new tools dedicated to the analysis of protein structural variability. These tools are now used to study a variety of biological processes,

A) Extraction and characterization of the n overlapping fragments of 4 $C\alpha$ 

B) Description of HMM-SA



C) Simplification of one 3D structure into one 1D sequence of structural letters

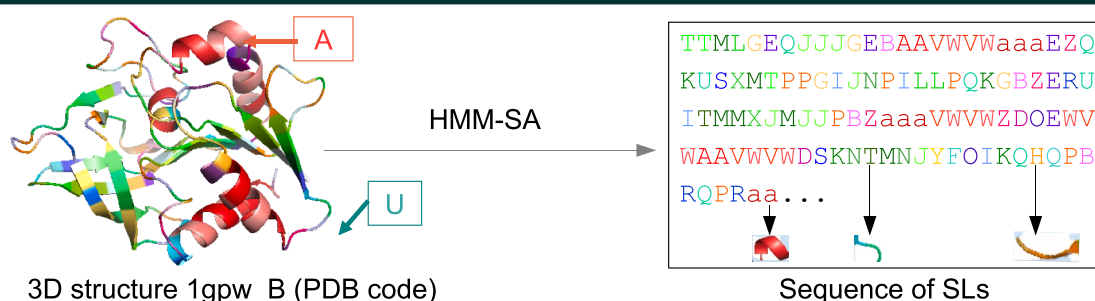


Fig. 1. Presentation of the HMM-SA structural alphabet. (A) Protein structures are divided into overlapping fragments of four alpha carbon atoms ($C\alpha$). The geometry of each fragment is described using three distances (d_1 , d_2 , d_3) between non-successive $C\alpha$ atoms and the projection d_4 of the fourth $C\alpha$ onto the plane formed by the other three. (B) Description of the HMM-SA structural alphabet. 1. Evolution of the Bayesian Information Criterion (BIC) as a function of the number of states in the structural alphabet. The analysis was performed on two independent datasets of 250 proteins each. For both datasets, a 27-state alphabet was identified as optimal. 2. Geometrical representation of the 27 structural letters of HMM-SA, labeled [A–Z; a]. The four letters outlined in red [A, a, V, W] correspond to structural motifs found in α -helices; the five letters outlined in green [L, M, N, T, X] correspond to β -strands; the remaining 18 letters describe loop conformations. 3. Transition matrix between the 27 structural letters. Letters are ordered from the most helical-like (a, A, V, W) to the most extended (L, N, M, T, X). The color of each cell indicates the transition probability between two letters, according to the scale on the right (from white for zero probability to red for probability 1). Highly colored areas (red/orange) indicate preferred local conformation transitions in protein structures, whereas white or pale cells represent rare or absent transitions. (C) Simplification of the 3D structure of protein 1gpw (chain B) into a sequence of structural letters using HMM-SA. The 3D structure and the structural letter sequence are both colored according to the 27-letter structural alphabet. 4. Matrix of the RMSD_{SL} between the 27 SLs. Letters are ordered from the most helical-like (a, A, V, W) to the most extended (L, N, M, T, X). The color scale indicates the magnitude of structural difference, with blue corresponding to low RMSD_{SL} values (i.e., similar conformations) and red indicating high RMSD_{SL} values (i.e., dissimilar conformations).

such as ligand-induced deformations and resistance mechanisms. In this review, we present these methodological advances and illustrate their applications through the structural analysis of HIV-2 protease and its interactions with inhibitors.

2. Introducing the HMM-SA structural alphabet

2.1. HMM-SA construction methodology

HMM-SA is a structural alphabet based on the description of protein conformations from the positions of C α atoms. HMM-SA was learned from a library of 1429 protein chains with less than 30 % sequence identity, extracted from the PDB and decomposed into four-residue fragments overlapping by three residues. The choice of four-residue fragments reflects a balance between capturing local structural features—such as backbone curvature and torsion—and keeping the alphabet size tractable. To learn the geometry of fragments, they were described by vectors of four descriptors corresponding to the three distances (d_1 , d_2 , d_3) between non-successive C α atoms and d_4 , the projection of the fourth C α onto the plane formed by the other three (Fig. 1A). To account for the sequential organization of fragments within protein structures during the learning process, a first-order Hidden Markov Model (HMM1) was employed. This memory-based model captures the dependency between consecutive structural letters, assuming a common underlying transition process across all polypeptide chains, governed by a first-order Markov chain. Structural alphabets of different sizes (R) were learned on two independent learning sets of proteins using HMM1 by progressively increasing R from 12 to 33 letters and compared using the Bayesian Information Criterion (BIC) [19], Fig. 1B. For a detailed description of the methodology used to build HMM-SA, including modeling with hidden Markov chains, please refer to the seminal works of Camproux et al. [8,15–17].

The optimal alphabet was obtained for 27 letters [8,15] (Fig. 1B). Each letter is described by its geometry and its transition probabilities with the other letters (Fig. 1B). The transition matrix obtained between the 27 structural letters is highly sparse: around 50 % of transitions are zero. This matrix reveals the preferential paths of local structures in protein structures and even in loops. The majority of structural letters exhibit low structural variability, indicating that each groups together fragments with similar local geometry [15]. To assess the structural similarity between the 27 letters, we computed their pairwise RMSD values (RMSD_{SL}) from 500 fragment pairs randomly chosen in the two structural letters [15] (Fig. 1B). The resulting matrix highlights groups of letters with similar backbone conformations (low RMSD_{SL}) as well as others with clearly distinct geometries (high RMSD_{SL}). A comparison between the 27 structural letters of HMM-SA and the regular secondary structures defined by the STRIDE program [20] showed that four letters were found to be specific to α -helices and five to β -strands. HMM-SA thus provides a detailed description of loops using 18 letters [15].

2.2. Encoding structures in HMM-SA space

HMM-SA aims to simplify the 3D structure of a protein into a series of structural letters where each letter corresponds to the local structure of a four-residue fragment (Fig. 1C). This encoding is based on both the local geometry of the fragments and the model's transition probabilities. This process, known as protein structure encoding, relies on Hidden Markov Model inference and can be performed using either the Viterbi algorithm, to obtain the most probable sequence of structural letters (MAP encoding), or the forward-backward algorithm, to compute posterior probabilities

for each letter at each position [21]. A high degree of concordance has been observed between the encoding produced by these two algorithms.

In 2016, two encoding improvements were proposed through the development of the SA-Flex structural alphabet [22]. In addition to the MAP encoding, SA-Flex introduces two encodings:

- NEFF (Number of Effective structural letters per position) encoding, which gives the actual number of structural letters at each position. NEFF coding reflects the effective number of structural letters compatible with the local geometry, based on the posterior probabilities computed from the Hidden Markov Model. Its value ranges from 1 (highly certain) to a theoretical maximum of 27, corresponding to a uniform distribution over all structural letters in HMM-SA. A low NEFF value indicates high certainty in the assignment: a single structural letter dominates the probability distribution, suggesting a well-defined local conformation. Conversely, higher NEFF values indicate ambiguity or local flexibility, where multiple structural letters have non-negligible probabilities. NEFF thus provides a measure of encoding uncertainty and local conformational variability.
- POST (*marginal posterior distribution*) encoding, which represents the distribution of structural letters at each position.

These different types of encoding (MAP, NEFF, POST) can be applied to any protein structure via the online tool available at <http://saflex.rpbs.univ-paris-diderot.fr/SA-Encoder.php>.

2.3. Comparison of HMM-SA with other structural alphabets

Unlike classical fragment libraries, structural alphabets aim to discretize protein 3D structure while preserving the sequential organization of local fragments. However, in most existing structural alphabets [9,12–14], this sequential dimension is not incorporated during the construction process but is instead analyzed separately. In contrast, HMM-SA integrates sequential dependencies during the learning phase itself, through the explicit use of a hidden Markov model. This framework allows the simultaneous modeling of both fragment geometry and their transition probabilities. In terms of fragment length, like the Conformational Letters [13] and M32K25 [14], HMM-SA is based on four-residue segments—an intermediate size offering a balance between local sensitivity and conformational stability. Other alphabets use variable fragment lengths: for instance, Protein Blocks rely on five-residue segments [9]. While Protein Blocks provide high-resolution encoding and have been widely used in structural analyses, the four-residue model implemented in HMM-SA emphasizes probabilistic encoding and transition modeling. This design supports efficient geometric discrimination and flexible applications, such as SA-Flex [22]. From a geometric standpoint, HMM-SA encodes fragment conformation using distance vectors between non-consecutive C α atoms. This representation captures non-local geometric features without relying on absolute coordinates or potentially redundant internal angles. In comparison, Protein Blocks use ϕ/ψ dihedral angle pairs for each residue [9], whereas M32K25 and Conformational Letters employ pseudo-angles derived from C α atom triplets [13,14]. In terms of conformational representation, HMM-SA comprises 27 structural letters, grouped into classes corresponding to major secondary structure families (helices, strands, and loops). This number exceeds that of more compact alphabets, such as the 16-letter Protein Blocks [9], and provides finer resolution, particularly in loop regions, which are often more structurally heterogeneous. Altogether, HMM-SA holds a distinctive position among existing structural alphabets.

It combines probabilistic modeling, robust geometric encoding, and explicit integration of fragment dependencies, making it a powerful tool for the analysis and modeling of protein conformations.

2.4. Methods and tools derived from HMM-SA

Given its ability to provide a detailed description of protein structures, HMM-SA has been extensively applied in studies focused on the analysis and classification of protein folds, as well as the investigation of side-chain conformations [23]. Furthermore, the simplification of 3D structures into 1D sequences has enabled the development of methods for rapid structural comparison [24], 3D peptide conformation generation [25,26], and the analysis of both intra- and inter-molecular protein interactions [27]. The HMM-based approach for constructing structural alphabet has also been successfully applied to porin proteins (a class of transmembrane proteins), leading to the definition of a porin-specific structural alphabet [28]. For a more comprehensive overview of these applications, readers are referred to reviews Regad et al., CSDA 2008; Reynès et al., 2011 [16,17] and the original publications.

With the increasing number of experimentally resolved protein structures available in the PDB, we have developed new methods based on HMM-SA, which are presented in the remainder of this review. We first introduce tools designed for protein functional annotation and structural motif prediction. We then describe recent developments aimed at characterizing structural deformations in protein targets, addressing growing demands driven by the expansion of structural databases and emerging challenges in drug discovery. Finally, we illustrate the application of these methods through a case study on the characterization of HIV-2 resistance mechanisms to protease inhibitors, demonstrating the potential of these approaches for understanding the underlying molecular mechanisms.

3. Functional annotation of proteins using HMM-SA and motif prediction

3.1. Predicting functional and structural motifs in protein loops

One of our first HMM-SA based projects involved the development of an approach to systematically explore the structural redundancy of protein loops by searching for recurrent structural motifs in these regions [29]. This method involves encoding 3D protein structures using HMM-SA, then extracting all 4-letter structural words (corresponding to 7-amino acid fragments) within the loop regions (Fig. 2). By analyzing these words for frequency, geometric and sequence conservation, we were able to create a bank of structural words corresponding to recurrent local conformations with high structural and sequence conservation. Searching for these structural words in a loop library revealed that, across both short and long loops, 73 % of the loop residues are part of fragments matching at least one structural word, demonstrating significant structural redundancy in these regions often considered as random. Structural word searching thus offers a simple and robust method for analyzing the structural diversity of loops on a large scale. The sequence and structural conservation of structural words suggest that they may play an important role in protein structure or function. To explore the potential link between structural words and biological function, we carried out a statistical characterization of structural words, inspired by approaches used in genomics. By analogy with these methods, we applied an approach based on over- or under-representation detection to the structural letter space of HMM-SA, in order to identify structural

words overrepresented in loops extracted from proteins belonging to the same SCOP superfamily (Structural Classification of Proteins [65]). This approach is based on the hypothesis that the exceptionality of words in proteins with the same function could reflect structural or functional specificity. However, classical statistical approaches used in genomics rely on valid approximations for long sequences and small alphabets, which is not appropriate for our context: loops are short fragments, coded using an alphabet of 27 letters. These differences lead to significant biases in the calculation of word statistics. To overcome this limitation, an exact method for calculating word statistics, based on the use of finite automata, was developed in collaboration with Dr. Grégory Nuel [30]. The advantage of this new method is that it requires no a priori assumptions about sequence distribution. Calculating the exact word statistics in the loops encoded by HMM-SA has led to the identification of functional words that are over-represented in a single SCOP family, many of which correspond to binding sites for major biological ligands: ATP/GTP, calcium, heme, NADP, SAM/SAH, iron, etc. (Fig. 2A). Searching for these structural and functional words within a 3D structure of a protein encoded in HMM-SA space provides functional and structural information on a protein (Fig. 2B). This annotation is accessible via the SA-Mot web server (<http://sa-mot.mti.univ-paris-diderot.fr>) [30–32]. The identification of structural and functional motifs within loops by the SA-Mot tool typically requires access to the 3D structure of the protein. However, for a large number of proteins, only the amino acid sequence is available. To overcome this limitation, a prediction method based on decision trees was developed by Dr. Christelle Reynès, allowing the prediction of structural words directly from sequence information (Fig. 2B). The method is presented in detail in Refs. [17,33]. This method was applied to investigate the structural basis of functional specificity in calcium-dependent protein kinases (CDPKs) from the model plant *Arabidopsis thaliana*. CDPKs are serine/threonine kinases found exclusively in plants and certain protozoa, characterized by calmodulin-like domains containing EF-hand motifs responsible for calcium binding. We combined biochemical sensitivity analyses of 14 CDPKs and the prediction of the calcium binding sites (EF-hand motif described by the structural word DODQ) from their amino-acid sequences using HMM-SA based method. These results showed that the absence of at least one EF-hand motif, or the alteration of two such motifs, in a CDPK sequence was sufficient to explain the reduced or abolished calcium sensitivity observed in some of the 14 isoforms [34]. This prediction strategy can in fact be generalized to any structural word of interest. In combination with available 3D structures or predicted models (e.g., from AlphaFold), this framework allows the annotation of entire proteomes with putative structural words and their associated functions, offering a powerful tool for functional inference and protein characterization at scale. This approach has already been successfully applied to an uncharacterized protein with known structure (PDB ID: 2RHM). SA-Mot detected specific structural words—associated with ATP/GTP-binding motifs—provided functional insights consistent with kinase activity, despite low sequence identity with annotated homologs [32]. This opens the door to proteome-wide analyses, where functional or structural motifs—previously defined in HMM-SA space—can be systematically predicted from primary sequences.

While these studies have focused primarily on loop regions and their involvement in function, similar principles can be applied to other structurally or functionally relevant regions of proteins. In particular, a promising direction is the analysis of highly flexible or even intrinsically disordered regions, to investigate whether specific HMM-SA letters or structural word patterns are enriched in such contexts. Similarly, functionally important regions such as

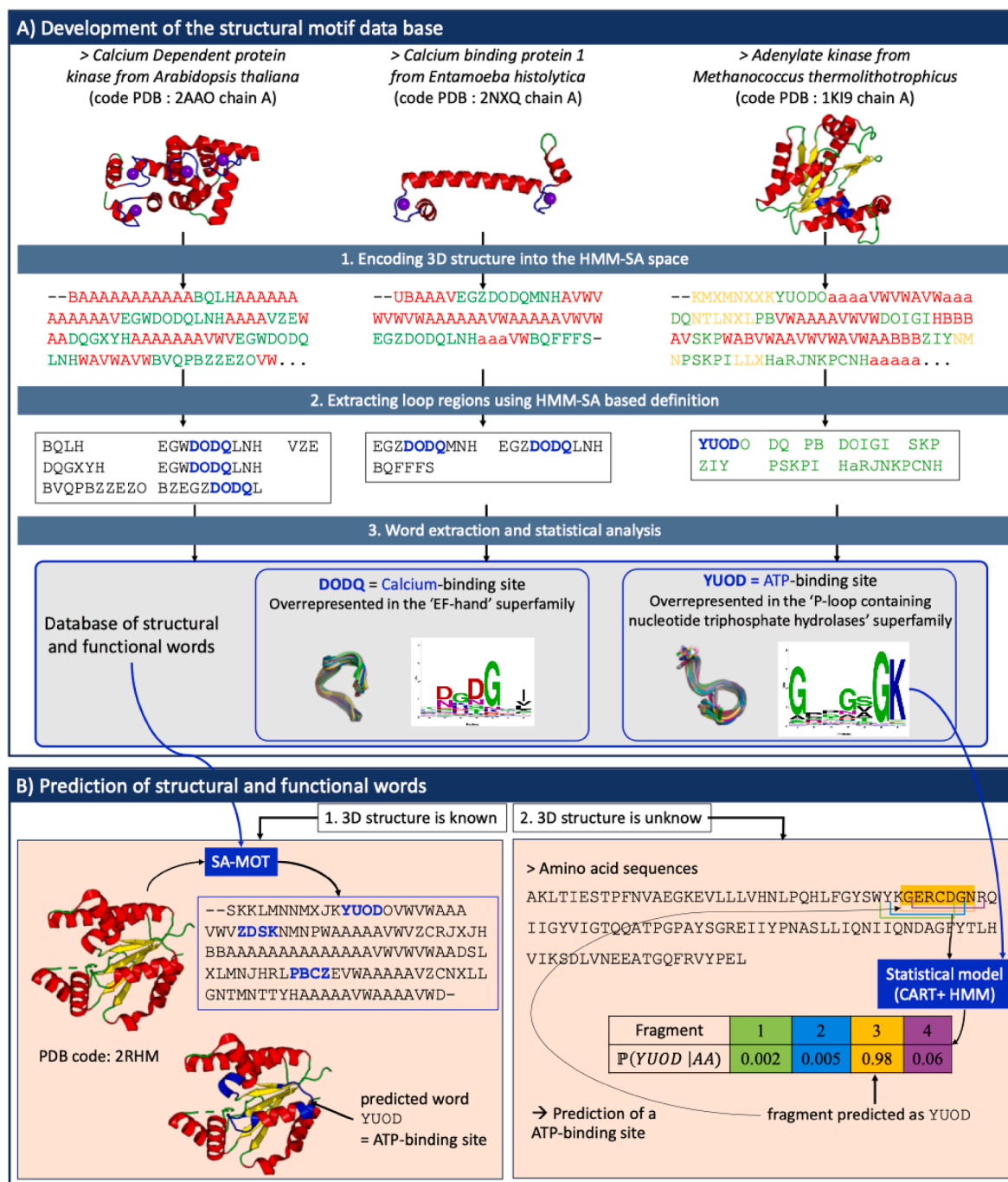


Fig. 2. Construction of the structural and functional word database (A) and structural/functional annotation of proteins through word matching (B). (A) The protocol for extracting structural and functional words comprises four main steps and takes as input the 3D structures of proteins. In step 1, input protein structures are simplified into 1D sequences of structural letters using the HMM-SA structural alphabet. The resulting sequences are colored according to their secondary structures: red for α -helices, yellow for β -strands, and green for loops. In step 2, loop regions are extracted from the structural letter sequences using a secondary structure definition based on HMM-SA. Recurrent 4-letter structural words (colored in blue) are extracted from these loop sequences. In step 3, the conservation in terms of amino acid sequence and geometry of recurrent words is assessed, and their statistical occurrences are computed within SCOP superfamilies [65]. A structural word is defined as a 4-letter word overrepresented across multiple SCOP superfamilies and is likely important for maintaining protein structure. A functional word is overrepresented in only one superfamily and is often associated with specific functional sites. Two examples of functional words are shown: DODQ, associated with calcium-binding sites, and YUOD, associated with ATP- or GTP-binding sites. For each word, structural alignments of the corresponding fragments and their sequence conservation are shown. This protocol led to the construction of a database containing 3310 structural and functional 4-letter words. Identifying these words in protein structures provides insights into regions critical for structural stability or function. (B) Two prediction strategies are available for identifying structural and functional words: 1. When the 3D structure of the protein is available. First the 3D structure is encoded into a sequence of structural letters using HMM-SA. This sequence is then scanned against the word database to identify matches and predict the presence of structural or functional words. This structure-based annotation is available via the SA-Mot webserver [<http://sa-mot.mti.univ-paris-diderot.fr>] [32]. The annotation of protein 2RHM (PDB code) is illustrated: based on its structural letter sequence, SA-Mot identifies two structural words (ZSDK and PBSZ) and one functional word (YUOD), which corresponds to an ATP-binding site. Thus, SA-Mot predicts two structurally important regions and one ATP-binding site. 2. When the 3D structure is not available, a prediction method based on the amino acid sequence has been developed [33]. A statistical model combining CART and HMMs is trained to predict the likelihood that a 7-residue fragment encodes a specific structural or functional word (e.g., YUOD). The input amino acid sequence of the target protein is fragmented into overlapping 7-residue windows, and the model estimates the probability of each fragment corresponding to the word of interest. High-probability fragments are predicted as occurrences of the target word, allowing the identification of potential functional sites such as ATP-binding regions.

ligand-binding pockets or protein–protein interfaces represent interesting targets for structural word profiling. For example, in the study by Martin et al. (2008), we examined the structural composition of protein interfaces across 68 complexes by encoding their conformations in HMM-SA space [35]. A statistical comparison of interface versus non-interface regions revealed distinctive conformational signatures: regular secondary structure letters (α -helices [A, a, V, W] and β -strands [L, M, N, T, X]) were under-represented in interfaces, whereas loop-associated letters were more frequent, though not uniformly so—certain loop-specific letters [U, F, P, K] were in fact under-represented. These results point to a heterogeneous and context-specific structural composition of interfaces, suggesting that local flexibility—as captured by structural letters—plays a role in mediating protein–protein interactions. Overall, these findings support the idea that distinct classes of structural words or letters may be associated with different levels of flexibility or structural constraint, depending on their functional context.

3.2. Sequence linkage and peptide structure prediction

Initial work on HMM-SA explored the distribution of amino acids in the 27 structural letters [36]. Using statistical criteria (Zscores, Shannon entropy [37]), it has been demonstrated that the 27 structural letters display varying levels of amino acid conservation, revealing particular amino acid preferences. Our work on exploring structural redundancy within protein loops has shown that this conservation in terms of amino acids was also observed for longer fragments of 7 residues [29]. This work thus demonstrates the ability of HMM-SA to efficiently encode sequence–structure relationships at different scales, paving the way for protein modeling suitable for large-scale structural prediction and annotation applications. In this context, HMM-SA is a powerful concept in *de novo* structure prediction of peptides and mini proteins, making peptide folding predictions faster and more computationally tractable. It has been successfully used for *de novo* structure prediction. This gave rise to the PEP-FOLD3 site, regularly used to predict peptides or small protein folding. These approaches rely on HMM-SA to model local fragment conformations and simulated annealing to optimize global structure by fragment assembly. Simulated annealing is a stochastic optimization algorithm that helps to find a conformation close to the energy minimum. The HMM-SA concept in PEP-FOLD (PEP-FOLD3 web server) is explained in several publications, including Efficient Sampling of Conformational Space. The HMM-SA approach enables the rapid generation of probable local 3D conformations from the amino acid sequence using statistical profiles trained on known structures [26,38]. By working with HMM-SA, the PEP-FOLD series bypasses the need for full atomic simulations during the early prediction stages [39,40]. HMM-SA remains efficient even when peptides are modeled in the context of molecular complexes or bound to partners [26].

The integration of HMM-SA into PEP-FOLD has enabled fast and reliable *de novo* prediction of short peptide structures, provided a solid foundation for rational peptide design, protein engineering, and therapeutic development.

4. Characterization of protein target structural deformations using HMM-SA

Proteins are flexible macromolecules whose dynamic properties play a key role in many biological processes. This flexibility conditions their ability to interact with different partners and is essential to their function. Predicting conformational changes induced by partner binding is therefore crucial for understanding

the molecular mechanisms underlying the biological functions of proteins, as well as for modeling protein–ligand complexes. In recent years, major advances in experimental resolution and structure prediction methods have led to an explosion in the number of structures available, both experimental and predictive. It is now possible to access numerous structures of the same protein, obtained under different experimental or biological conditions (in the presence or absence of a ligand, ions, mutations, etc.). Systematic comparison of these structural ensembles offers a unique opportunity to explore the diversity of conformations adopted by a protein, and thus to characterize structural deformations associated with functional events. Traditionally, structural comparisons are made after optimal superposition of the structures. However, this approach may be limited in the case of localized deformations or intrinsic flexibility. We have previously shown that HMM-SA provides a powerful tool for quantifying structural redundancy within protein loops, by identifying regions of local similarity without the need for global superposition. On this basis, we have explored the relevance of HMM-SA-based approaches for identifying and characterizing structural deformations induced by various factors such as partner binding, mutations, or changes in experimental conditions.

4.1. Characterization of the structural deformation induced by the binding of a protein partner

We used HMM-SA to localize structural deformations induced by protein partner binding. The analysis was based on a comparison of the sequences of structural letters associated with the same protein in its bound and unbound states: a change of letter at a given position reflects a local conformational deformation induced by the interaction. This comparison revealed that 36 % of structural letters undergo at least one modification, indicating that protein–protein binding generates structural rearrangements detectable by HMM-SA (Fig. 3A). The analysis showed that local deformations are significantly more frequent at interfaces than in other regions of the structure, underlining the direct impact of partner binding on the conformation of interface regions. To characterize in more detail deformation induced by partner-binding, we then analyzed the resulting conformation after binding, namely the structural letter substitutions. We established a substitution probability matrix considering only substitutions associated with local RMSD_{SL} values greater than 0.2 Å to focus on meaningful conformational changes (Fig. 3B). This matrix highlights preferred conformational transitions, revealing that structural deformations are not random: some structural letters are more prone to switch into others, particularly within the same secondary structure group. Substitutions often occur within helices or strands, while transitions between secondary structure types are less frequent.

Using a similar approach to the search for patterns in loops, we highlighted recurring structural motifs (or "words" in HMM-SA letters) that were conserved across different protein interfaces in their bound forms, while the corresponding regions adopted distinct local conformations in the unbound states. Fig. 3A illustrates this observation using two targets analyzed in Ref. [35]: the murine monoclonal antibody 5G9 (anti-human tissue factor) and pig pancreatic α -amylase. The comparison of structural letter sequences of unbound and bound structures of both targets revealed local conformational changes upon binding, particularly at the interface. Notably, both complexes exhibited a shared structural motif (GOIF) at the interface in the bound form, despite differences in the corresponding unbound forms. In the antibody, the structural deformation was minor (GOIJ $\rightarrow$ GOIF; RMSD_{unbound/bound} = 0.4 Å), suggesting a subtle adaptation upon

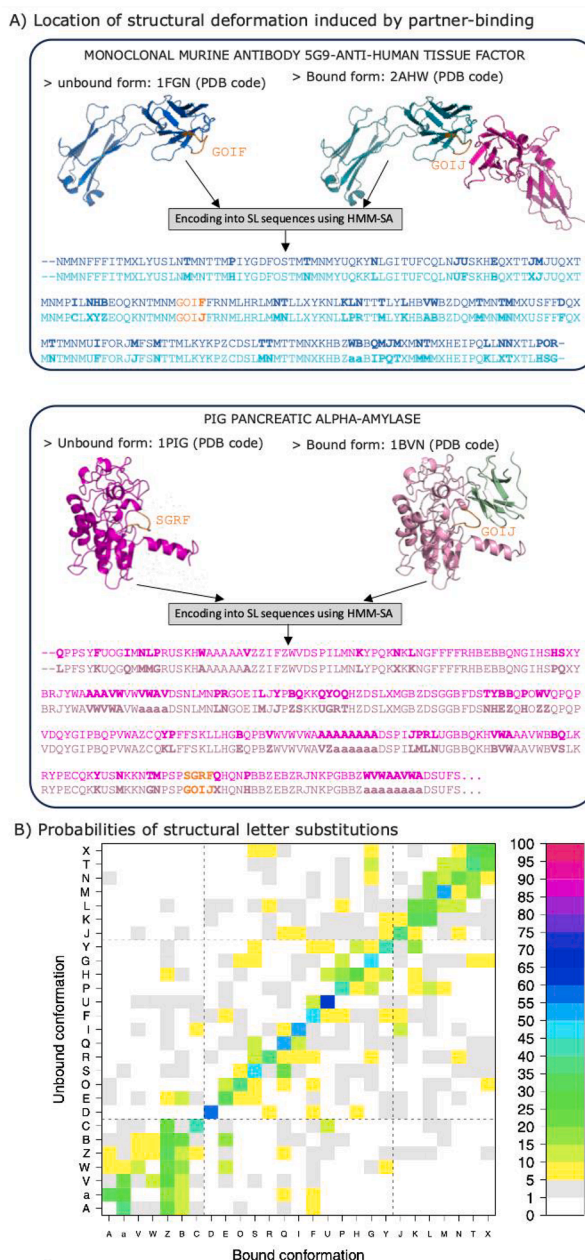


Fig. 3. Structural rearrangements upon partner binding in two protein targets. A) Analysis of the conformational changes induced by partner binding in the murine monoclonal antibody 5G9 (anti-human tissue factor) and in pig pancreatic α -amylase. For each target, the unbound and bound structures were encoded into structural letter sequences using HMM-SA. The structural letter sequences of the apo and holo forms were then compared to identify changes in local backbone conformation upon binding. In the structural letter sequences, bold letters indicate positions where the structural letters differ between the apo and holo forms. Regions highlighted in orange correspond to structural motifs adopting the same conformation in the bound forms of both proteins, potentially reflecting conserved binding-related structural patterns. B) Probabilities of structural letter substitution associated with local RMSD_{SL} greater than 0.2 Å in the interface set. The unbound form is taken as the reference for the computation of substitution probabilities.

antigen binding. In contrast, α -amylase underwent a more pronounced deformation (SGRF $\rightarrow$ GOIF; RMSD_{unbound/bound} = 0.8 Å) upon partner interaction. The recurrence of the GOIF motif in the two unrelated complexes suggests a common conformational signature associated with partner binding. This example shows how HMM-SA can reveal structurally conserved interface motifs beyond sequence similarity. This suggests the existence of conformational signatures common to interaction processes, paving the way for future applications in interface recognition, functional annotation or molecular partner design.

4.2. SA-conf: a tool for quantifying the structural variability of a protein target - interest in studying its different conformations

Previous studies have shown that HMM-SA is a powerful tool for localizing and quantifying structural deformations induced by the binding of a protein partner [35]. Following on from this work, we wanted to explore the ability of HMM-SA to detect other types of structural deformations, such as those induced by the binding of small molecules, by amino acid changes, or by variations linked to experimental conditions. This approach is based on the

comparison of local structures of the same protein resolved in different contexts—whether in the presence of different ligands, under different experimental conditions, or carrying mutations. Thanks to advances in structural resolution techniques—such as X-ray crystallography, cryo-electron microscopy, and NMR—the number of entries in the PDB continues to grow, with increasing redundancy due to multiple structures of the same proteins solved under such varying conditions. Thus, by coupling the creation of adequate datasets, it becomes possible to systematically analyze the structural deformations associated with each of these factors. However, there are few tools available for comparing a set of protein structures. These tools, such as MUSTANG [41] and MulPBA [42], produce an alignment of structures, but provide little information on target variability. To address this need, we developed a new tool, named SA-conf (Structural Alphabet conformations) based on HMM-SA, specifically designed to localize and quantify both sequence and structural variability within a target protein for which multiple structures are available [43]. Fig. 4 summarizes the main computational steps of the SA-conf workflow. SA-conf takes as input a list of protein structure identifiers. It first calculates a multiple alignment of the amino acid sequences extracted from the submitted 3D structures. SA-conf then simplifies the structures into structural letter sequences. This data is then used to convert the sequence alignment into structural alignment. To do this, the amino acids in the sequence alignment are replaced by the corresponding structural letters. From the two alignments (sequence and structural letters), SA-conf calculates the conservation of each position in terms of amino acids and structural letters, using a criterion based on Shannon entropy [37]: neq_{AA} for amino acids and neq_{SL} for structural letters (Fig. 4). Thus, a position with a neq_{SL} (resp. neq_{AA}) close to 1, i.e. where a single structural letter (resp. amino acid) is observed in all structures, will be considered structurally (resp. in terms of sequence) conserved (Fig. 4). Conversely, a position with several structural letters (resp. amino acids) ($neq_{SL} > 1.5$) (resp. amino acids: $neq_{AA} > 1.5$) will be defined as structurally (resp. in terms of sequence) variable. This parameter quantifies the number of structural letters observed at each position, thereby enabling the detection of residues that exhibit variability either in sequence, in local structure, or both. In addition, SA-conf determines the number of secondary structure categories observed at each aligned position, as determined using the HMM-SA definition [44]. This measure serves as an indicator of the conformational variability amplitude across the structural ensemble. A structurally variable position where multiple structural letters from the same regular secondary structure class (e.g., α -helices or β -strands) are observed is generally considered to exhibit low-amplitude variability. In contrast, positions where structural letters from distinct secondary structure classes co-occur (e.g., helix and loop, strand and loop, or α -helices and β -strands) reflect more substantial conformational changes. In addition, this deformation can be more precisely quantified by the user using the pre-computed RMSD values between structural letters (RMSD_{SL}, Fig. 1B). A change between two structural letters with a high RMSD_{SL} reflects a more substantial deformation. As expected, the most significant structural changes generally occur between α -helix and β -strand letters, while the smallest occur between letters belonging to the same secondary structure class.

Cross-analyzing variability in terms of sequences and structures provides insights into the relationship between amino acid changes and conformational deformation in the target. In addition, the study of target structural variability detected by SA-conf as a function of biological knowledge of the target provides information on factors that may explain the structural changes observed: substitutions/mutations, intrinsic flexibility of the target, partner binding, experimental artifacts ... One of the original features of

our tool is that it can be applied to structures obtained by all kinds of techniques, such as crystallographic structures, those obtained by nuclear magnetic resonance, those extracted from molecular dynamics simulations or even structures obtained by homology modeling.

SA-conf was applied to characterize the structural variability of several protein targets: the HIV-1 protease, the catalytic domain of human urokinase and p53 [43]. The three protein systems were selected based on their therapeutic relevance and the availability of rich and diverse structural datasets. HIV-1 protease is a widely studied enzyme involved in viral maturation, with numerous structures resolved by both X-ray crystallography and NMR. Urokinase is a serine protease implicated in tumor invasion and metastasis, with a large number of X-ray structures available in complex with different inhibitors. Finally, the tumor suppressor p53 is a key regulatory protein for which extensive structural data are available, including X-ray structures of various mutants complexed with different partners and homology models. These datasets were chosen to showcase the versatility of SA-conf in analyzing structures derived from different experimental methods or modeling approaches, and in detecting both intrinsic conformational variability and structural changes induced by mutations or partner binding. These illustrations confirmed that SA-conf is an effective and rapid tool for detecting structural distortions induced by the binding of a partner, by substitutions, or resulting from the natural flexibility of the protein. In the cases of p53 and urokinase, the regions with high structural variation as detected by SA-conf frequently overlapped with known functional domains or mutation hotspots reported in the literature. The comparison of urokinase structures complexed with different ligands enabled the identification of ligand-induced conformational changes. This result demonstrates that SA-conf is capable of capturing structural variability specifically associated with ligand binding. This approach was further applied to investigate similar ligand-induced deformations in other therapeutically relevant targets, including the HIV-2 protease (PR2) [45], a key enzyme in viral maturation and a validated target for antiretroviral therapy, and the anti-apoptotic protein Bcl-xL [66], which plays a central role in cancer cell survival and is targeted by pro-apoptotic small-molecule inhibitors. These analyses revealed that most of the structural deformations detected by SA-conf are located at ligand-binding sites, and predominantly affect rigid residues. This supports the idea that the observed variability is not due to intrinsic flexibility but rather reflects external factors such as ligand binding. Interestingly, we also identified conformational changes in rigid residues located outside the binding site, suggesting potential allosteric effects induced by ligand interaction. For a detailed analysis of the structural variability of the PR2 binding site, see subsection 5.2. Together, these results reinforce the conclusion that the deformations captured by HMM-SA are not merely structural fluctuations, but reflect biologically meaningful variability associated with functional or regulatory roles. These illustrations confirmed that SA-conf is an effective and rapid tool for detecting structural deformations induced by partner binding, substitutions or as a result of the protein's natural flexibility. The SA-conf software executable is openly available at: <https://owncloud.rpbs.univ-paris-diderot.fr/owncloud/index.php/s/uOdItX27jUfgQM6>.

4.3. Protocol for locating and quantifying structural asymmetry in an oligomer

Building on previous results that demonstrated the ability of HMM-SA to capture local structural variations induced by experimental conditions, partner binding or amino-acid changes, we

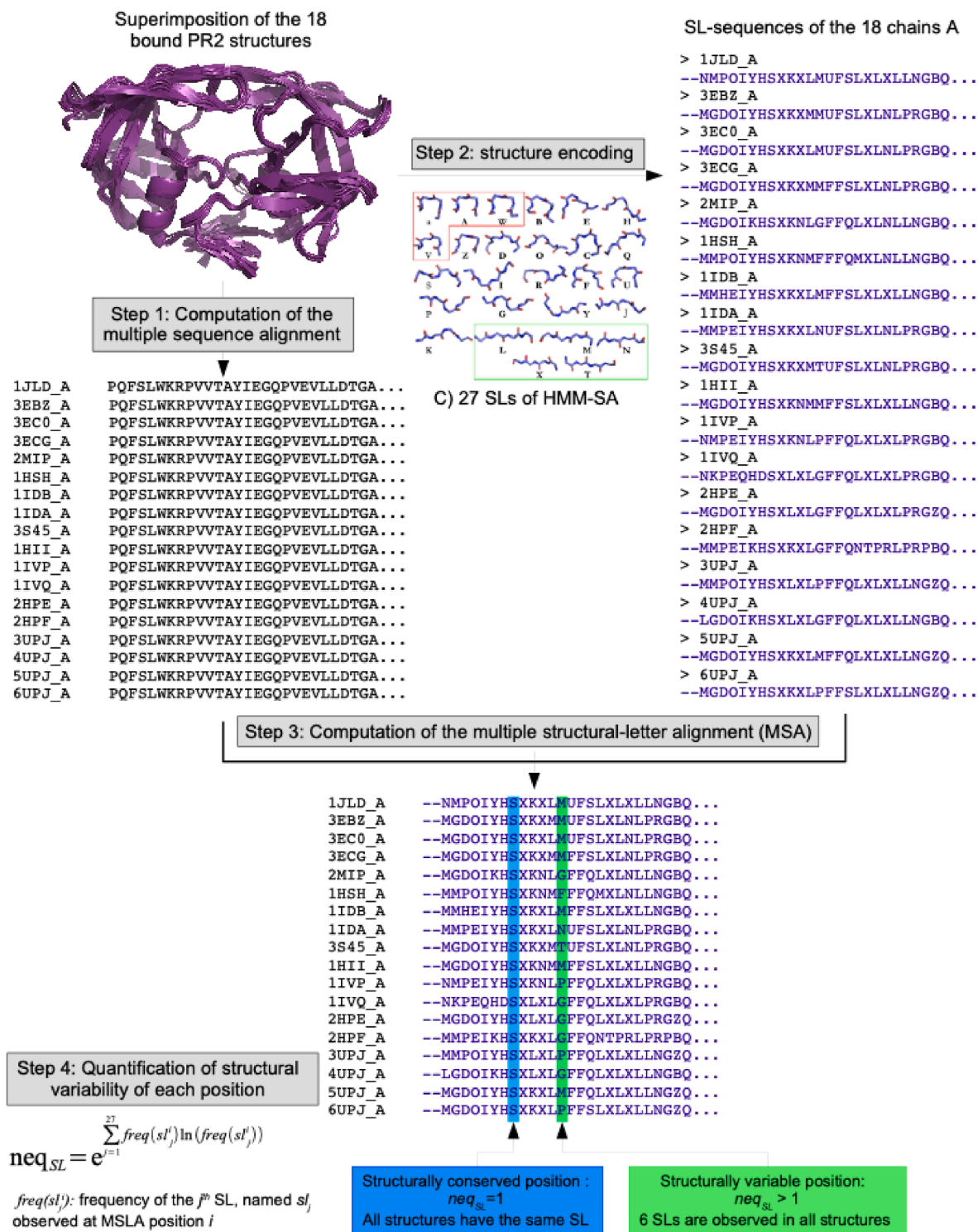


Fig. 4. Overview of the SA-conf protocol [43] illustrated on the HIV-2 protease. SA-conf takes as input a list of 18 PDB codes corresponding to different structures of HIV-2 protease. Step 1 consists in extracting the amino acid sequences from all the PDB files and computing the multiple sequence alignment of these sequences. Step 2 consists in simplifying the 3D structures into sequences of structural letters using the HMM-SA alphabet, where each letter encodes the local conformation of a residue. In Step 3, SA-conf generates a structural alignment based on the multiple sequence alignment and the structural encodings. Step 4 quantifies sequence and structural variability at each alignment position using the exponential of Shannon entropy (noted neq_{AA} for amino acids and neq_{SL} for structural letters). A position with neq_{SL} (or neq_{AA}) equal to 1 corresponds to a position that exhibits the same structural letter (or amino acid) in all structures, indicating a structurally (or amino-acid) conserved position. Conversely, a high neq_{SL} (or neq_{AA}) value reflects a position with multiple structural letters (or amino acids), indicating structural (or sequence) variability.

applied this approach to the characterization of structural asymmetry in homo-oligomeric protein assemblies. Homo-oligomeric protein structures are composed of multiple chains sharing

identical amino acid sequences. While these chains are expected to adopt similar conformations, comparisons of chain conformations often reveal orientation differences, resulting in partially

asymmetric structures (Fig. 5). This structural asymmetry observed in crystallographic structures can result from a variety of factors: experimental conditions [46], the intrinsic properties of the protein, or the binding of various ligands (partner proteins, ligands, substrates) [47,48]. Structural asymmetry plays an essential functional role, notably by contributing to the specificity of interaction with different molecular partners [47–50]. The detection and quantification of asymmetry in a homo-oligomer relies on the structural comparison of chains of identical sequence, for example by calculating the RMSD between pairs of chains (Fig. 5) [35,43]. Given the ability of HMM-SA to efficiently compare local conformations, we have developed an original protocol for detecting and quantifying the structural asymmetry of a homo-oligomer based on this tool. This method aims to identify positions showing local conformational variations between chains of the same structure. The protocol includes two main steps (Fig. 5):

- Step1: encoding each chain as a sequence of structural letters (from HMM-SA), each letter representing the local conformation of a residue,
- Step 2: comparison of structural letters between chains for each aligned position.

A position is said to be symmetric if all chains have the same structural letter at that position, indicating a locally conserved conformation (Fig. 5). On the other hand, a position is defined as asymmetric if at least one of the chains has a different structural letter, reflecting a local deformation of the backbone (Fig. 5). Similarly to the SA-conf approach, the amplitude of conformational changes can be assessed using two complementary parameters. It can be interpreted by examining the secondary structure classes of the differing structural letters between the two chains (Fig. 1B), as well as the RMSD_{SL} values between those letters (Fig. 1B). Together, these indicators help distinguish minor local rearrangements from more substantial conformational deformations. Our method offers a simple, rapid and robust approach to locating asymmetric positions in the protein backbone. Unlike other methods, it requires neither superposition nor calculation of angles or distances and enables direct comparison of the local conformations of the backbone of different chains. Depending on the input structure analyzed with the HMM-SA-based approach, different types of asymmetry may be detected. For instance, when the structure corresponds to an apo crystallographic form, asymmetry may arise from crystal packing, intrinsic flexibility, or interchain interactions. In contrast, when analyzing a structure bound to a ligand or a protein partner, additional asymmetries can emerge due to ligand-induced conformational changes. To distinguish between these possible sources of asymmetry, the analysis can be complemented by examining residue-level flexibility using B-factor values. Residues identified as asymmetric and displaying high B-factors are more likely to reflect local flexibility or dynamic fluctuations. Conversely, asymmetric residues with low B-factors (rigid residues) indicate deformations that may result from external factors such as ligand binding, crystal packing, or oligomeric interface constraints. Further insights can be gained by mapping asymmetric positions onto functional regions of the protein, such as ligand-binding pockets, protein-protein interfaces, or crystal contact zones. This spatial distribution helps to interpret the structural origin of the observed asymmetry and assess its potential functional relevance. Fig. 5 presents an analysis of structural asymmetry in the crystallographic structure 1AHS (PDB code), corresponding to the N-terminal region of the VP7

protein from the African horse sickness virus. VP7, a key component of the intermediate viral capsid, plays a crucial role in viral assembly and particle stability. The structure includes a trimeric form of the domain, in which the three monomers were compared pairwise (AB, AC, BC) using the HMM-SA-based asymmetry detection protocol. Each pair exhibits between 24 and 27 asymmetric residues, broadly distributed across the structure except in the N-terminal region. Most of these asymmetric positions involve low-amplitude structural deformations; however, some correspond to higher-amplitude changes. More than 30% of the changes occurs between β -strand structural letters (Fig. 5). On average, a structural letter change occurs with a mean RMSD_{SL} of 0.67 ± 0.33 Å, with the highest RMSD_{SL} observed between G and F structural letters (1.7 Å). Most of these asymmetric residues are not conserved across all comparisons, revealing local and pair-specific conformational differences within the trimer. This suggests that the association between monomers within the trimer is not strictly symmetrical, and that each interface adopts distinct local conformations. Only four positions (176, 177, 198, and 240) are asymmetric in all three comparisons. Among these, positions 198 and 240 are located in the central region of the trimeric interface, indicating that their asymmetry is driven by trimeric interface interactions and likely reflects interchain constraints. In contrast, positions 176 and 177 are located in a solvent-exposed loop, where asymmetry is more likely due to local flexibility.

This study highlights the ability of the HMM-SA approach to detect fine structural asymmetries within multimeric assemblies. Such analyses can reveal flexible regions of a protein target and fine mechanisms of conformational adaptation upon partner binding. Consequently, the detection of asymmetric positions may help identify residues involved in structural deformation and serve as markers of structural, functional, or allosteric constraints within oligomers. This type of analysis provides valuable insights into protein assembly, conformational dynamics, and function.

5. Application to the study of HIV-2 resistance to protease inhibitors

In this last section, we illustrate the relevance of HMM-SA in therapeutic target analysis through the case study of HIV-2 protease (PR2), a key enzyme in antiretroviral drug development. HIV-1 and HIV-2 are the two viruses responsible for AIDS. Although these viruses have distinct origins, they share the same mode of infection and replication cycle. However, HIV-2 is less transmissible than HIV-1, and the period between HIV-2 infection and the development of AIDS is considerably longer than that observed with HIV-1. HIV-1 and HIV-2 infections are treated by a common therapeutic arsenal, consisting of antiretroviral molecules that target the four main mechanisms of the viral replication cycle: (i) fusion inhibitors, which prevent the viral genome from entering the cytoplasm of the target cell by blocking the fusion of viral and cellular membranes (ii) reverse transcriptase inhibitors, which prevent the reverse transcription of viral RNA into DNA that can be integrated into the genome of the infected cell (iii) integrase inhibitors, which block the integration of viral DNA into the host cell genome, and (iv) protease inhibitors, which prevent the maturation and assembly of viral proteins. Some of these antiretrovirals, initially developed against HIV-1, are not effective against HIV-2. Indeed, HIV-2 is naturally resistant to non-nucleoside reverse transcriptase inhibitors, fusion inhibitors and certain protease inhibitors [51–54]. In addition to this natural resistance, HIV-2 has also developed acquired resistance through mutations that enable it to resist inhibitors. It has been shown that a single mutation in

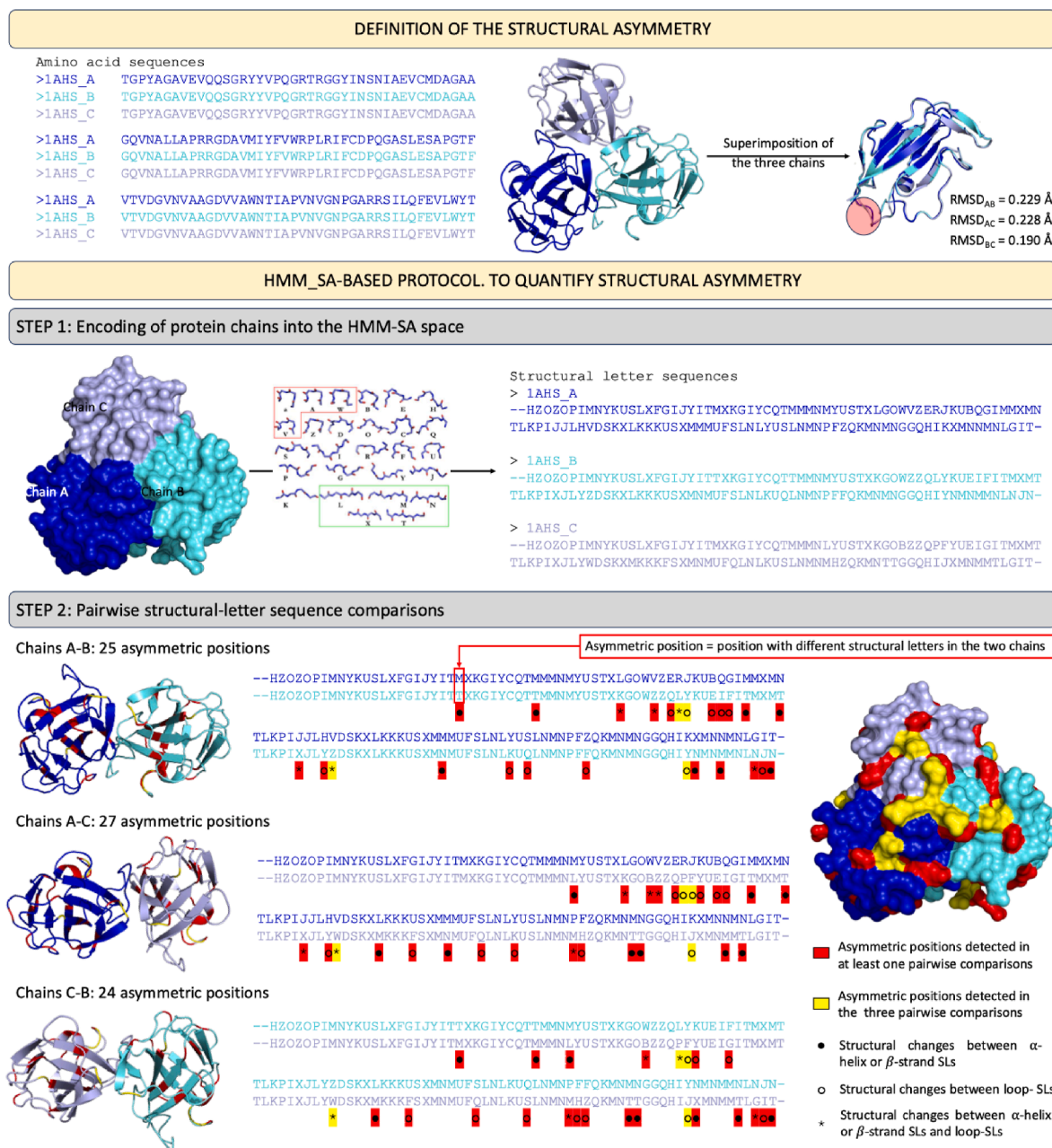


Fig. 5. Detection of structural asymmetry in a protein structure. (A) Illustration of structural asymmetry identified in protein 1AHS (PDB code). The 1AHS structure contains three chains with identical amino acid sequences. Comparison of their conformations after optimal structural superposition and pairwise RMSD calculation reveals local structural differences between the three chains. (B) Protocol for detecting structural asymmetry based on HMM-SA. This protocol takes as input the 3D structure of a protein (here, 1AHS) and involves two main steps. In Step 1, each chain is simplified into a sequence of structural letters using HMM-SA. In Step 2, the structural letter sequences of each chain are compared pairwise to identify asymmetric positions, i.e. positions with different structural letters in the two compared chains. The total number of asymmetric positions across the structure provides a quantitative measure of global structural asymmetry. The 3D structure of protein 1AHS is shown in cartoon (left) and surface (right) representations, with each chain colored distinctly. Asymmetric positions are highlighted in red, and those identified as asymmetric in all three pairwise comparisons (chains A-B, A-C, and B-C) are shown in yellow. Positions of asymmetry involving structural letters from different secondary structure classes are marked with a "*", those involving structural letters α -helix or β -strands SLs are marked with a "•", and those involving structural letters from α -helix or β -strands SLs and loop-SLs are marked with a "o".

the HIV-2 protease can confer resistance to all protease inhibitors [55,56]. The mechanisms of acquired resistance of PR2 remain poorly understood. Moreover, no in-depth structural studies of PR2 have been carried out to understand the impact of these resistance mutations. In collaboration with Pr Diane Descamps and Dr. Benoît Visseaux, we have carried out a precise and comparative structural study of wild-type and mutant forms of PR2 to provide information to better understand the mechanisms of acquired resistance. In particular, a detailed understanding of ligand-

induced deformations is essential to decipher the molecular mechanisms governing PR2 recognition and inhibition, as well as the impact of substitutions between HIV-1 protease (PR1) and PR2 on the protease structure. Indeed, ligand-induced deformations can directly affect the geometry and dynamics of the active site, influencing both the binding affinity and the inhibitory efficacy of antiretroviral drugs. In the case of PR2, certain mutations associated with drug resistance may not prevent inhibitor binding entirely, but instead alter the induced fit or dynamic behavior of

the enzyme upon ligand interaction. Understanding these structural rearrangements helps clarify how PR2 accommodates or resists specific inhibitors, and how its response differs from that of PR1. This knowledge is key for rational drug design, as it enables the identification of structural features that contribute to resistance and the development of molecules better adapted to the conformational landscape of PR2. In order to investigate how these ligand-induced and mutation-driven deformations manifest structurally in PR2, we designed a three-part strategy combining multiple HMM-SA-based tools. First, we applied the HMM-SA asymmetry detection protocol to a set of crystallographic PR2 structures in both ligand-free and ligand-bound forms to identify and quantify natural and ligand-induced asymmetries. Second, we used SA-conf to compare 19 crystallographic structures of PR2 complexed with different inhibitors, allowing us to detect conserved and variable structural positions across drug-bound states. These two complementary approaches together reveal the intrinsic flexibility of PR2 and the local conformational adaptations that occur upon ligand recognition. Third, we performed a comparative analysis of molecular dynamics structures of PR1 and PR2 using SA-conf [43], enabling us to explore the impact of sequence substitutions on structural dynamics and flexibility, particularly around the inhibitor binding site. This integrative strategy provides a detailed view of the structural landscape of HIV proteases and highlights the relevance of HMM-SA-based methods for characterizing conformational signatures linked to drug resistance. Together, these results provide key structural insights into how mutations and ligand binding shape PR2 conformational response, helping to clarify the structural features underlying inhibitor selectivity and resistance.

5.1. Study of PR2 structural asymmetry

Structural asymmetry analysis is a relevant approach to explore local deformations within homodimers, particularly those induced by ligand binding [47,48]. In PR1, it has been shown that the structural asymmetry plays a role in the specificity of PR1 for its substrates [49,50]. However, no studies have yet described the structural asymmetry of PR2 or its role in the recognition of various inhibitors. Such an analysis could provide valuable insights into the molecular mechanisms underlying the natural resistance of PR2 to certain inhibitors. To address this objective, we applied the HMM-SA-based protocol for localizing and quantifying the natural asymmetry in the structure of ligand-free PR2 (PDB code: 1HSI) [57]. This analysis revealed that 35 % of positions exhibit different local conformations (Fig. 6). The 1HSI structure is a crystallographic structure of unbound PR2. Thus, the detected variability may arise from the intrinsic flexibility of PR2 or from crystal packing. To assess whether the detected asymmetry has true biological significance or is merely the result of crystallographic artifacts, we compared the structural asymmetry of the unbound PR2 structure before and after energy minimization. This approach allows us to remove crystal contacts while preserving biologically meaningful conformations. Several asymmetric positions observed in the crystallographic structure were no longer asymmetric after minimization—particularly those located near residues involved in crystal contacts ($C\alpha$ atoms within 4.5 Å of symmetry mates)—indicating that part of the asymmetry can indeed be attributed to packing effects. However, a substantial number of asymmetric positions (80%) persisted after minimization. This supports the conclusion that these asymmetric conformations reflect intrinsic properties of the protein, such as local flexibility or functional structural rearrangements. To explore in more detail the deformation, we characterized the structural letter pairs between chains A and B. Most of these combinations (71%)

appeared only once, while a few—such as L–M—were recurrent across multiple sites. Interestingly, the majority of variations occurred between closely related structural letters, suggesting subtle local adjustments rather than large conformational changes.

In a second step, we compared the structural asymmetry of the 19 PR2 crystallographic structures available in the PDB (18 ligand-complexed structures and one ligand-free structure) to explore the structural variability of PR2 induced by ligand binding [58]. We showed that 31% of PR2 positions are asymmetric in at least one structure, and that on average one position was detected as asymmetric in six structures (± 4.6). On average, an asymmetric position exhibits a moderate magnitude, quantified by an RMSD_{SL} of 0.6 Å (± 0.28). However, 12% of asymmetric positions displayed high-magnitude asymmetry (RMSD_{SL} greater than 1 Å). Conversely, 33% of positions showed low-magnitude asymmetry with RMSD_{SL} below 0.5 Å. To assess whether the structural asymmetry observed in PR2 could result from specific experimental conditions or crystal packing artifacts, we performed a series of complementary analyses. We found no significant correlation between structural resolution and the number of asymmetric positions (Pearson correlation coefficient = -0.15). Moreover, structures crystallized in the same space group (e.g., $P2_12_12_1$) showed highly variable levels of asymmetry, suggesting that space group symmetry alone does not determine the extent of asymmetry. These results indicate that the observed structural asymmetry cannot be fully explained by experimental conditions, and may instead reflect intrinsic or functionally relevant conformational variability. Given the heterogeneity of the PR2 dataset, the observed structural variability likely reflects a combination of intrinsic flexibility and external factors, including, crystal packing effects, and ligand-induced conformational changes.

To better understand how intrinsic flexibility contributes to the structural variability observed across the PR2 structures, we analyzed variability in relation to residue flexibility, quantified by the B-factor. High B-factors generally indicate flexible or mobile regions, while low B-factors suggest well-ordered and potentially functionally constrained sites. The combination of B-factor profiling with HMM-SA asymmetry detection revealed that some asymmetric positions occur in highly flexible regions, whereas others correspond to rigid areas likely explained by external factors, such as ligand binding. Mapping the asymmetric residues onto the PR2 structure revealed that 42% are located at the dimer interface (Fig. 6), suggesting that part of the asymmetry arises from deformations caused by monomer-monomer interactions. Comparison of the asymmetric positions detected in the apo and ligand-complexed forms highlighted 53 bound-specific asymmetric positions (Fig. 6), meaning they are asymmetric in at least one bound PR2 dimer but not in the unbound dimer. Among them, 15% of these positions are located in the binding pocket (Fig. 6), suggesting that the structural asymmetry observed at these sites is a direct consequence of ligand binding. Analysis of the occurrence of these asymmetric positions in relation to the type of bound ligand revealed a potential link between structural asymmetry and ligand specificity. In particular, certain asymmetric positions appear to be specific to ligands, suggesting they may play a role in ligand recognition. These results highlight the existence of a natural structural asymmetry in PR2 that can also be induced by the interaction of the two monomers and the binding of different ligands.

To further investigate the nature of asymmetric positions, we examined whether certain local backbone conformations are more specific to asymmetric positions. By analyzing the distribution of the 27 structural letters at asymmetric *versus* symmetric positions, we observed notable trends that shed light on the relationship

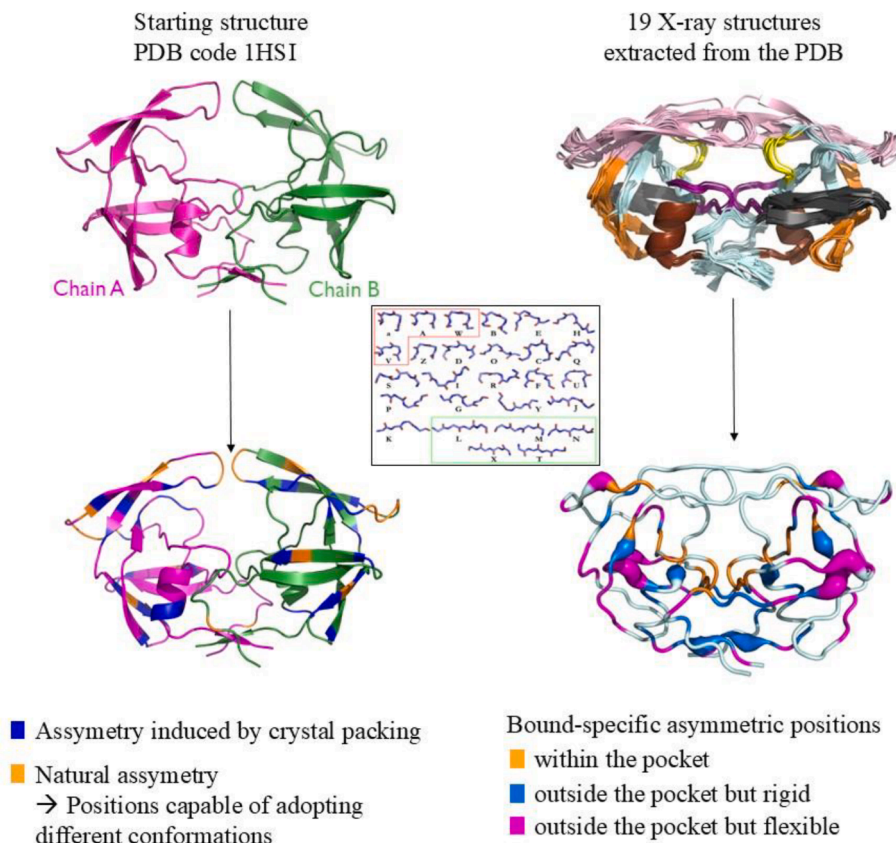


Fig. 6. Localization of structural asymmetry in PR2 structures. A) Mapping of asymmetric positions observed in the apo form of PR2 (PDB code: 1HSI). The structure is shown in cartoon representation, with chain A colored in magenta and chain B in green. Asymmetric positions were identified using the HMM-SA-based protocol, which detects differences in local conformation between the two chains. Two categories of asymmetry are distinguished: positions with crystal packing-induced asymmetry (colored in blue), considered as experimental artifacts, and positions with asymmetry attributed to intrinsic protein flexibility (colored in orange). B) Localization of bound-specific asymmetric positions. These positions were identified by analyzing and comparing the structural asymmetries across 19 crystallographic structures of PR2 in complex with various ligands. All structures were superimposed onto the apo form (1HSI) for consistency. Representations are shown in cartoon mode and colored according to the main structural and functional regions of PR2: the elbow-flap region (light pink), the wall (yellow), the fulcrum (orange), the catalytic site (purple), and the α -helix (brown). Bound-specific asymmetric positions—i.e. residues that are asymmetric in the ligand-bound forms but symmetric in the apo form—are mapped onto the 3EBZ structure. This structure is colored in sky blue and displayed in putty cartoon mode where large diameter indicates the overrepresented asymmetric positions. Bound-specific asymmetric positions are color-coded according to their spatial proximity to the binding site: orange for those within the binding pocket, light blue for those outside the pocket but structurally rigid (based on B-factors), and magenta for those outside the pocket and structurally flexible.

between conformation type and asymmetry. Our results reveal that some conformations, structural letters *a* and *v* are over-represented in the asymmetric positions, while structural letters *B*, *D*, and *T* are underrepresented. These findings highlight a possible relationship between local backbone geometry and the emergence of asymmetric structural behavior in PR2. This suggests that certain local conformations may be more prone to structural perturbations, potentially contributing to asymmetry.

The approach based on HMM-SA proved particularly relevant for detecting, locating and quantifying structural asymmetry. More specifically, it enables us to distinguish deformations due to the protein's intrinsic flexibility from those induced by external factors, such as crystal packing or inhibitor binding. The identification of asymmetric positions specific to certain ligands suggests a functional link between structural asymmetry and molecular recognition, reinforcing the interest of such analysis to better understand the mechanisms of ligand-protein interaction and guide the development of new targeted inhibitors.

5.2. Study of structural deformations of PR2 induced by inhibitor binding: exploration of the binding specificity of different inhibitors

In a second study, we sought to locate and precisely

characterize the structural deformations undergone by PR2 upon binding to different inhibitors, with the aim of gaining a better understanding of the molecular mechanisms of recognition and structural adaptation involved in the inhibition of PR2. To this end, we compared the local conformations of each protein position in all 19 crystallographic structures of PR2 available in the PDB (apo and ligand-complexed forms), using the SA-conf tool [59]. This analysis of the local structures of the 19 PR2 structures, revealed that a set of positions (23 % of positions) adopt an identical local conformation across all 18 complexed structures. This conservation suggests that these structurally conserved residues play an essential role, whether in catalytic activity, ligand recognition or maintenance of the protein's overall architecture. SA-conf also revealed that 77 % of PR2 residues adopt different local conformations in at least one of the structures (Fig. 7), highlighting a high degree of structural variability in PR2. According to the composition of the dataset, this variability may be linked to the intrinsic flexibility, experimental conditions, or the effect of ligand binding. By studying the flexibility of structurally variable residues—estimated using the average B-factor computed across the 19 crystallographic PR2 structures—we were able to distinguish deformations resulting from the protein's intrinsic dynamics from those specifically induced by inhibitor binding. In particular,

deformations observed in flexible residues are likely due to intrinsic structural fluctuations, while those occurring in rigid residues suggest variability caused by external factors such as ligand binding. Interestingly, both rigid and structurally variable positions are observed within the binding site as well as outside of it. This finding suggests that ligand binding induces deformations beyond the binding site, highlighting the presence of allosteric structural changes. By coupling structural variability analysis with the analysis of inhibitor-residue interactions, we established a detailed cartography of the binding pocket. This analysis revealed two main categories of residues within the PR2 binding site. The first include conserved conformations across complexes that consistently interact with all inhibitors, indicating a central role in ligand anchoring. The second comprise residues showing ligand-dependent structural variability that often form interactions specific to certain ligands, and are likely involved in the specificity of inhibitor recognition (Fig. 7B). These residues cluster into distinct regions: the pocket floor, which includes the catalytic site, is structurally conserved and interacts with conserved ligand moieties, whereas the entrance and left side of the pocket show greater flexibility and contribute to ligand-specific recognition and selectivity (Fig. 7B). Interestingly, this analysis also revealed that certain structural deformations occur recurrently in response to specific ligand chemotypes, suggesting that some local conformational changes may be preferentially induced by distinct molecular scaffolds. For example, three PR2 structures complexed with inhibitors chemically related to 4-hydroxycoumarin (PDB codes: 3UPJ, 5UPJ, 6UPJ) display the same structural letter at position 84A, distinct from that observed in other complexes. This suggests a consistent backbone deformation induced by this chemical group. Similarly, comparing the local structures of PR2 bound to amprenavir and darunavir—two chemically related inhibitors with markedly different binding affinities—revealed structural divergences at positions 23B, 31A, 47A, and 48A. These results reveal specific regions of PR2 with notable structural plasticity that adapt their conformation in response to ligand binding. Such ligand-induced conformational signatures suggest a

role in modulating molecular recognition. In future studies, investigating the relationship between these structural deformations and inhibitor activity could help identify conformational signatures associated with high-affinity binding. These results could be exploited in drug design to identify compounds that stabilize favorable PR2 states, thereby enhancing affinity and selectivity. The conformational fingerprints revealed by SA-conf thus offer valuable insights for the rational development of next-generation PR2 inhibitors.

Comparative analysis of local conformations using the SA-conf tool thus enabled us to identify a set of structural residues essential for PR2 stability and interaction with inhibitors, as well as to locate residues undergoing deformation after ligand binding, highlighting regions involved in the binding specificities of PR2 inhibitors. These findings not only provide deeper insights into the conformational adaptation mechanisms of PR2 to various inhibitors but also help identify key structural determinants underlying binding specificity. Moreover, they offer valuable clues into the natural resistance mechanisms of PR2, highlighting how local conformational changes can interfere with inhibitor binding.

5.3. Study of PR2 innate resistance mechanisms: exploration of the impact of substitutions between PR1 and PR2 on the local structure of PR2

After analyzing the structural deformations induced by the binding of various protease inhibitors to gain insight into their binding specificity, we explored the impact of amino acid substitutions between PR1 and PR2 on their structural conformations (Fig. 8). This approach aims to identify substitution-induced structural variations that could modulate inhibitor binding or affect PR2 dynamics, thereby influencing recognition and binding. To this end, we used SA-conf to compare local conformations in PR1 and PR2 structures generated through molecular dynamics simulations, which enabled us to account for the intrinsic flexibility of both proteases. Molecular dynamics simulations were performed on the ligand-free forms of PR1 and PR2 (PDB codes:

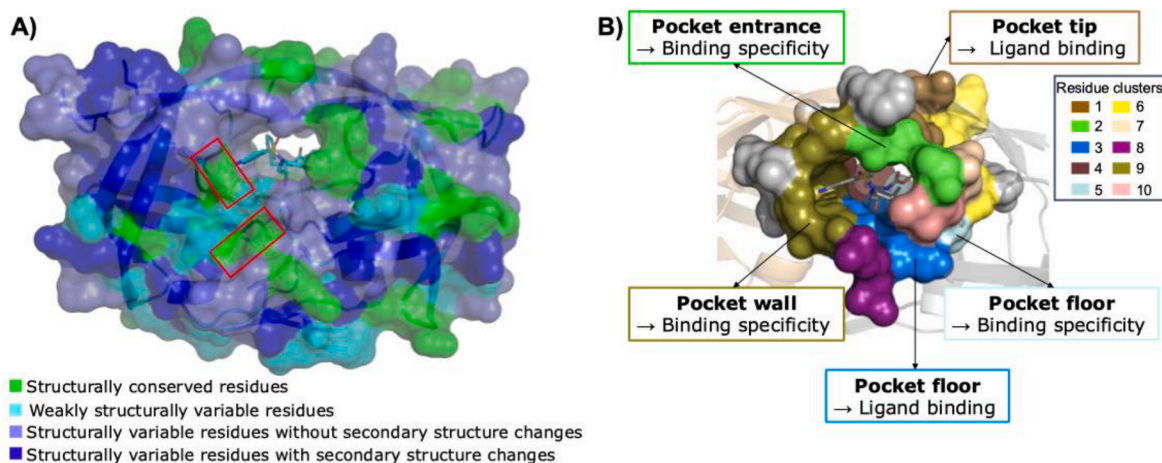


Fig. 7. Structural variability characterization of PR2 based on the comparison of the 19 crystallographic structures available in the PDB. (A) Structural variability of PR2 assessed by comparing local conformations across the 19 structures using the SA-conf program. Different types of positions (structurally conserved or variable) are mapped onto the PR2 structure (PDB code: 3S45), shown in surface and cartoon representations, and colored according to residue-level structural variability. Green residues correspond to positions with conserved local conformations across all 19 structures. Cyan residues are weakly variable, showing different local conformations in only a few structures. Blue residues are highly variable, but maintain the same secondary structure in all conformers. Dark blue residues display both diverse local conformations and different secondary structure assignments across structures. The ligand (amprenavir) is shown as sticks and colored cyan. Red highlights indicate regions with conserved structure that appear critical for maintaining the ligand-binding site. (B) Characterization of the binding site based on the structural variability and conservation of interactions between inhibitors and PR2. The binding site is shown in surface representation and colored according to clusters grouping residues establishing the same interactions with inhibitors. Regions establishing numerous conserved contacts with ligands are defined as important for ligand binding affinity, while regions establishing ligand-specific contacts are defined as implicated in binding specificity.

1HHP and 1HSI, respectively), using GROMACS 2022 [60] and the AMBER99SB force field [61]. Each simulation was run for 500 ns, and frames were extracted every nanosecond, resulting in two datasets of 501 frames each. The detailed simulation protocol is provided in [Appendix 1](#). SA-conf was then applied to these datasets, with chains A and B treated separately. [Fig. 8B](#) shows the neq_{SL} values calculated for 98 positions across 501 frames extracted from the PR1 and PR2 molecular dynamics simulations. Overall, both proteases display similar neq_{SL} profiles for chains A and B, with no significant difference in average neq_{SL} values (Kruskal-Wallis test, $p = 0.87$). Positions 7, 28, 29, 45, 68, 81, and 82, for example, adopt identical conformations in PR1 and PR2 and remain highly conserved throughout the simulations. Conversely, amino acid substitutions between PR1 and PR2 appear to influence structural variability: certain residues, including those in the catalytic site, the elbow region, and the α -helix, exhibit greater flexibility in PR2 compared to PR1. Conversely, residues such as 36, 40, and 41 show reduced variability in PR2. At position 40, for instance, the structural letter [J] is conserved in over 86 % of chain A frames in PR1, while in PR2, the most conserved letter is [Y], present in only 51 % of frames. Notably, differences in structural letter distributions are observed both at substituted and non-substituted residues, indicating that mutations can induce allosteric effects that extend beyond the mutation sites themselves. Our results show that substitutions directly impact the local conformation and flexibility of the inhibitor binding site, the α -helix region important for binding site conformation, and the elbow region involved in binding site closure after inhibitor

binding. These conformational alterations can modulate the accessibility, dynamics and stability of the active site, thus affecting inhibition efficacy.

To further investigate whether mutation-induced structural deformations detected by SA-conf are biologically relevant, we analyzed a previously characterized mutant of PR1 (referred to as PR1_M) bearing the V32I, I47V, and V82I substitutions. Tie et al. (2012) solved the crystal structure of this mutant in complex with the inhibitor amprenavir (APV) [62]. These mutations allow PR1_M to adopt a binding site geometry that mimics that of PR2. We applied SA-conf to compare the local conformations in three structures: PR1-APV (PDB code: 3EKV), PR1_M-APV (PDB code: 3S43), and PR2-APV (PDB code: 3S45). The analysis revealed that 49 % of positions have a neq_{SL} value of 1, indicating identical conformations across the three proteases. Importantly, ten positions (50, 51, 75, 81, 83, 87, 89, 90, 93, and 94) adopt the same structural letters in PR1_M and PR2 but differ from PR1, suggesting that the observed conformational changes are indeed induced by the introduced mutations. Notably, positions 50, 81, and 83 are located near the mutated residues I47V and V82I, reinforcing the link between local structural deformation and specific substitutions. These results confirm that the deformations captured by SA-conf are functionally relevant and reflect mutation-induced adaptations. All these results show that HMM-SA is powerful for detecting and quantifying mutation-induced deformations, offering a robust framework for the fine analysis of resistance mechanisms.

A) Amino acid sequences

PR1 PQTILWCRPIVITIKIGGQKEALLDTGADITVLEEMSLPGRWPKMIGGGIGGFIKVRQYDQILIEICGHKAIGTVLVGTFVNIIGRNLLTQIGCTLNF
PR2 PQFSLWCRPVVTAYIEGQFVEVLLDTGADISIVAGIELGNNYSPIKVGIGGGFINTKEYKNVEIEVLNKKVRATIMTGTPINIFGRNILTALGMSLNL

B) neq_{SL} values



C) Structural letter sequences

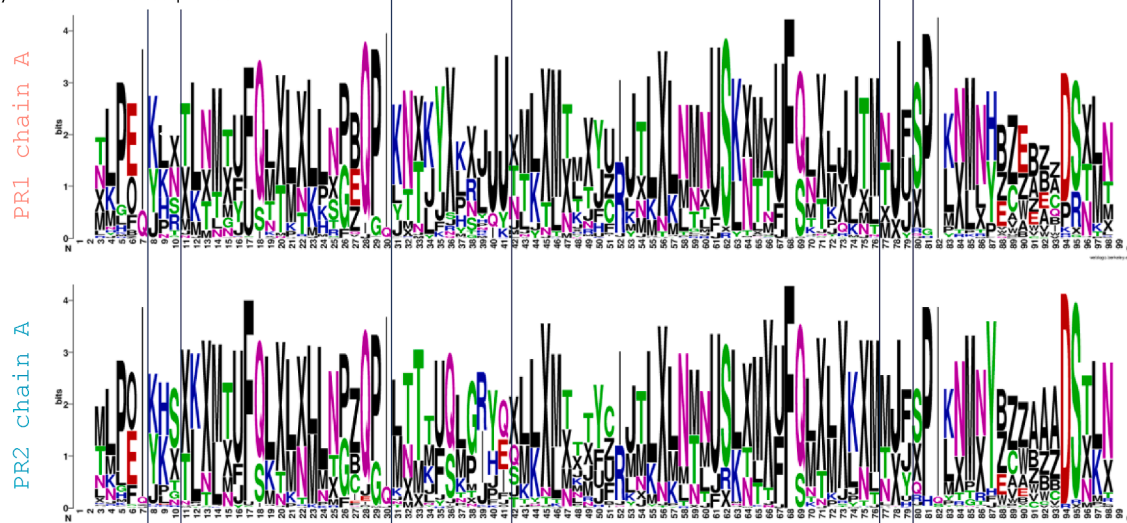


Fig. 8. Impact of substitutions between PR1 and PR2 by coupling molecular dynamics simulations and the SA-conf tool. (A) Amino acid sequences of PR1 (in light red) and PR2 (in light blue). Purple residues indicate amino acid substitutions between the PR1 and PR2 sequences. (B) Values of neq_{SL} for the 99 positions of chain A in PR1 and PR2, calculated from the 1000 structures extracted from molecular dynamics simulations of 1HHP (PR1 apo) and 1HSI (PR2 apo) using the SA-conf tool. (C) Structural letter logos of chain A for PR1 and PR2, generated using the encoding into HMM-SA space of the 1000 structures obtained from the molecular dynamics simulations. Each column corresponds to a PR position, and the stacked letters represent the relative frequency of each structural letter observed at that position. The height of each letter indicates its frequency, while the total stack height reflects the overall structural conservation at that position, expressed in bits. Logos were generated using WebLogo [63].

6. Conclusion and outlook

Recent advances in deep learning, particularly AlphaFold2 [18], have transformed structural biology by fulfilling Anfinsen's hypothesis—that a protein's sequence encodes its 3D structure—while overcoming the combinatorial bottleneck of the Levinthal paradox. These tools now provide highly accurate static models of protein folds directly from amino acid sequences. However, understanding how protein structures vary across functional states, respond to ligand binding, or adapt to mutations remains a major challenge. Such structural plasticity plays a central role in many biological mechanisms, including allosteric regulation, dynamic switching, and drug resistance. Predictive models often fail to capture this conformational diversity or to offer interpretable insights into local flexibility. To complement these models, high-resolution tools capable of systematically quantifying local deformations across structure ensembles are still essential.

The structural alphabets initiated by Prof. S. Hazout represent a major methodological advance for the description, comparison and analysis of protein structures. HMM-SA is distinguished by its use of a hidden Markov model, which allows the modeling of transitions between structural letters [8,15]. This approach captures sequential dependencies and improves the structural encoding of protein conformations. This modeling results in a detailed description in 27 structural letters, that groups fragments exhibiting the same local geometry. By transforming complex 3D information into 1D sequences of structural letters, HMM-SA enables efficient, fine-grained exploration of local conformational variations, either of functional, dynamic or mutational origin. The applications presented in this review highlight the usefulness of HMM-SA for the functional annotation of structures [29,32,34,44], the characterization of ligand- or partner-induced deformations [35,45], and the quantification of asymmetry in oligomeric structures [58]. In particular, its use in the study of PR2 resistance to inhibitors illustrates the finesse of the analyses enabled by HMM-SA, highlighting deformations induced by inhibitor binding or those induced by substitutions between PR1 and PR2. The results we obtained have enabled us to identify specific structural variations that may alter inhibitor recognition or binding, and thus contribute to our understanding of the molecular mechanisms behind PR2 natural resistance. Additional applications of HMM-SA tools further support their generalizability beyond HIV-2 protease. In particular, Regad et al. demonstrated the effectiveness of these methods on structurally and functionally diverse proteins such as the tumor suppressor p53 and human urokinase-type plasminogen activator [43]. Despite their differing topologies and dynamic properties, HMM-SA successfully captured conformational variations linked to functional states or disease-related mutations in both cases. More recently, we applied SA-conf to the anti-apoptotic protein Bcl-xL, a key cancer therapy target, analyzing 130 chains from 60 X-ray structures [66]. This analysis enabled us to detect subtle ligand-induced deformations, separate them from mutation-driven effects, and generate a structural map of the binding site. Our findings show that SA-conf can capture both stabilizing and destabilizing effects of ligand binding. These results underscore the versatility of HMM-SA-based tools for identifying meaningful structural variability across diverse therapeutic targets.

Although most applications presented here were based on crystallographic structures, HMM-SA is equally suitable for analyzing dynamic datasets, including molecular dynamics simulations. Our analysis of molecular dynamics trajectories for PR1 and PR2 showed how HMM-SA can quantify flexibility and detect subtle conformational differences. Moreover, because it requires

only $\text{C}\alpha$ coordinates, HMM-SA is fully compatible with AI-predicted models, such as those from AlphaFold2. Confidence scores (e.g., pLDDT) can be used to filter uncertain regions, enabling robust post-analyses such as the detection of conserved motifs or structural diversity across AlphaFold-predicted ensembles.

The wealth of tools derived from structural alphabets, and the diversity of their applications, make them a valuable resource for computational structural biology. At a time when the volume of structural data is exploding, approaches such as HMM-SA appear indispensable for extracting relevant biological knowledge and exploring the quality of available data. The development of new tools based on this representation should continue to enrich our understanding of protein structure-function relationships and facilitate the rational design of innovative therapeutic solutions.

CRedit authorship contribution statement

Anne-Claude Camproux: Writing – review & editing, Visualization, Methodology, Formal analysis, Conceptualization. **Marine Baillif:** Writing – review & editing, Visualization, Software, Formal analysis, Data curation, Conceptualization. **Léa Dufay:** Writing – review & editing, Visualization, Formal analysis, Data curation. **Leslie Regad:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Project administration, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix I

Molecular dynamics simulations were carried out using the apo forms of PR1 and PR2. In the Protein Data Bank, a single unliganded PR2 crystallographic structure is available (PDB code: 1HSI), while several unliganded and wild-type PR1 structures are listed. We selected the structure corresponding to PDB file 1HHP as the starting point for PR1. As with all unbound PR1 crystallographic structures, the PDB file contains only one monomer. In order to reconstruct the dimeric form of PR1 in its apo state, we generated the second monomer by symmetric reflection from the structure of the first, using PyMOL's symmetry mates function, with a proximity threshold of 5 Å.

The two systems were prepared and simulated following the same protocol. Protonation states of ionizable residues were assigned at pH 7 using the PropKa 3 package [1]. In particular, the oxygen atom (OD1) of Asp25 in chain B was protonated according to PropKa predictions. Simulations were carried out using GRO-MACS 2022 [2] and the AMBER99SB force field [3]. Each protein was solvated in a cubic box of explicit TIP3P water molecules, with a minimum distance of 1.0 nm between the protein and the box edges. The systems were neutralized by adding Na^+ and Cl^- ions. All bonds involving hydrogen atoms were restrained using the LINCS algorithm. Long-range electrostatic interactions were computed using the Particle Mesh Ewald (PME) method with a grid spacing of 0.16 nm and an interpolation order of 4. van der Waals interactions were treated with a 1.0 nm cut-off. Non-bonded interactions were handled using the Verlet cut-off scheme, with a cut-off distance of 1.0 nm. Energy minimization was first performed during 50,000 cycles of steepest descent algorithm that was dismissed when the overall force was lower than 1000 kJ/mol/nm. Equilibration was conducted in two consecutive phases of 1 ns

each, using a 2 fs time step and position restraints. The first phase was run under the NVT ensemble, followed by equilibration in the NPT ensemble. Temperature was maintained at 310 K using the V-rescale thermostat, and pressure was controlled at 1 bar using the Parrinello-Rahman barostat. Finally, 1 μ s production simulations were carried out under NPT conditions, with a 2 fs time step.

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